

Learning Resource On Speech and Video Processing

Module-1: Speech Processing Concepts Chapter-1: Introduction to Speech Processing

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Chapter Outcomes

- After the completion of this module, the students will be able to:
 - Define Speech Processing
 - Identify the need of speech processing
 - Distinguish between speech transmission and speech processing
 - Explain speech processing model
 - List the application areas of speech processing
 - Identify the elements of speech communication
 - Describe speech production and perception systems
 - Explain acoustic phonetic representation of speech sounds
 - Classify and Explain different types of phonemes

Introduction to Speech Processing

- **Speech:** Speech is the natural mode of communication among human beings. Usually speech contains speaker information, language information, and message.
 - It is the natural process of producing and transmitting spoken language through sound waves to communicate information, ideas, and/or emotions.
- **Processing:** Processing is the systematic analysis, transformation, or manipulation of data or signals to extract useful information or to achieve a desired outcome.
- **Why Speech Processing???**
 - **Speech processing** is required to efficiently analyze, enhance, compress, recognize, and synthesize speech, enabling intelligent voice-based communication between humans and machines.

- “**Speech processing** is the study of speech signals and the processing methods of these signals. It involves **analysis, synthesis, recognition, and coding** of speech for applications in communication and human–computer interaction.”
 - *from “Fundamentals of Speech Recognition” by Lawrence Rabiner & Biing-Hwang Juang (Prentice Hall, 1993)*
- “Digital speech processing refers to **the representation, transformation, and interpretation** of speech signals using digital techniques, enabling compression, transmission, and recognition.”
 - *from “Digital Processing of Speech Signals” by L. R. Rabiner & R. W. Schafer (Prentice Hall, 1978)*

Speech Transmission vs Speech Processing

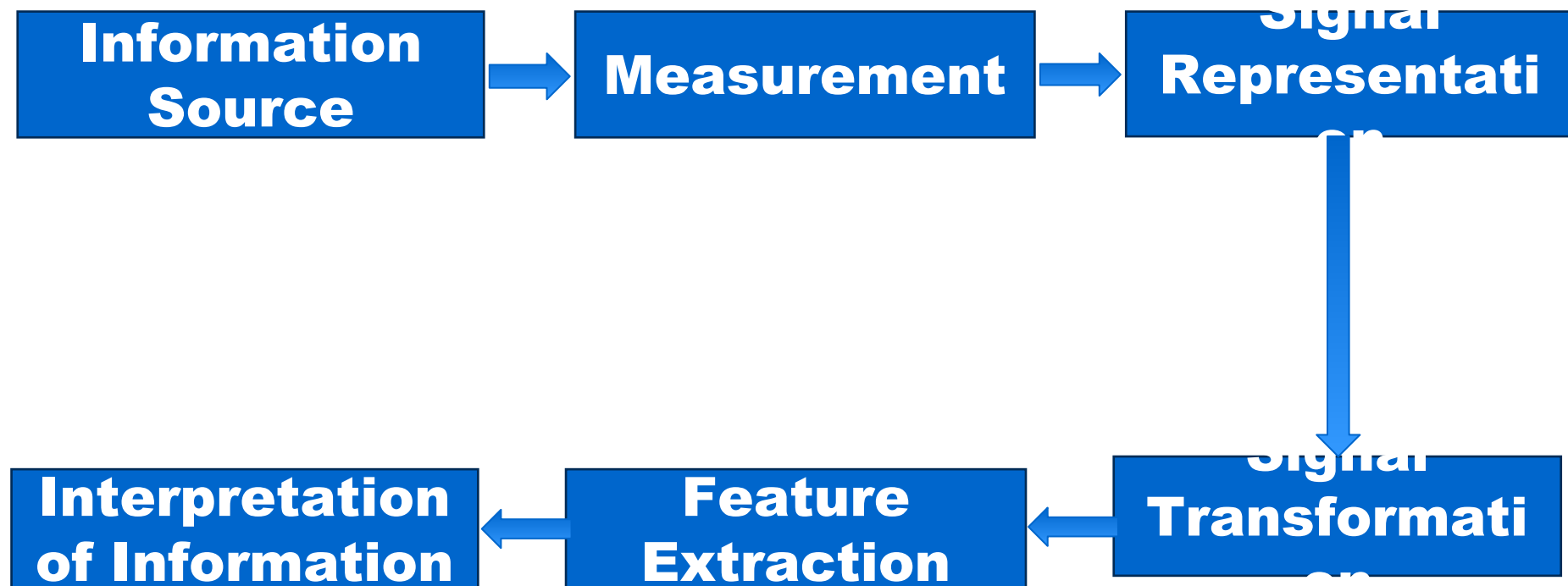
Speech Transmission: (Talking over mobile phone)

- **Analog/Digital Transmission** - Your voice is captured by the microphone, converted into electrical signals, and transmitted over cellular networks.
- **Compression & Encoding** - Uses codecs (like AMR, Opus) to compress audio for efficient bandwidth usage.
- **Network Transport** - Signals are carried over GSM, VoLTE, or VoIP protocols.
- **Goal:** Deliver your voice clearly to the other person, not analyze or interpret it.

Digital Speech Processing

- **Feature Extraction** - Converts raw audio into numerical features (MFCC, LPC).
- **Recognition & Interpretation** - Identifies phonemes, words, and meaning.
- **Applications** - Voice assistants, transcription, biometrics, translation.
- **Goal**: Understand, analyze, or act on speech content.

Speech Processing Model



- **Information Source:** The source that produces speech using some linguistic medium.
- **Measurement:** Capturing raw audio through microphones.
- **Signal Representation:** Discretization (Continuous signal to discrete signal), Quantization, digital encoding of speech signals.
- **Signal Transformation:** Noise reduction, filtering, normalization, framing, windowing, and speech enhancement.
- **Feature Extraction:** Converting waveforms into numerical features (e.g., MFCC, LPC).
- **Interpretation of Information:** Identifying phonemes, words, or speakers.

Applications of Digital Speech Processing

- **Smart Homes:** Voice assistants (Alexa, Siri, Google Assistant) control appliances, lighting, and entertainment systems.
- **Education:** Real-time captioning (transcription generation), pronunciation and fluency checks in language learning apps.
- **Healthcare:** Doctor-patient conversation, prescription assistance, speech aid devices
- **Media:** Live captioning for broadcasting and streaming platforms.
- **Business/Commerce:** Automated customer service through speech-driven chatbots
- **Automotive:** Voice commands for navigation, infotainment, and hands-free communication
- **Security:** Voice biometric authentication

Real-time Speech Processing Platforms

Deepgram: Provides ultra-low latency speech-to-text with advanced features like speaker diarization, sentiment analysis, and topic detection.

AssemblyAI: Offers real-time transcription with speaker labeling and scalable APIs for voice applications.

OpenAI Realtime: Combines speech recognition with GPT-powered conversational AI for multilingual interactive assistants.

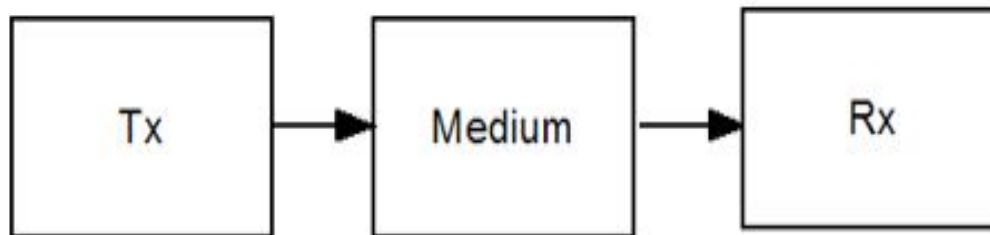
Google Cloud Speech-to-Text: Enterprise-grade speech recognition supporting over 125 languages with high accuracy and cloud integration.

Elements of Speech Communication

Human:



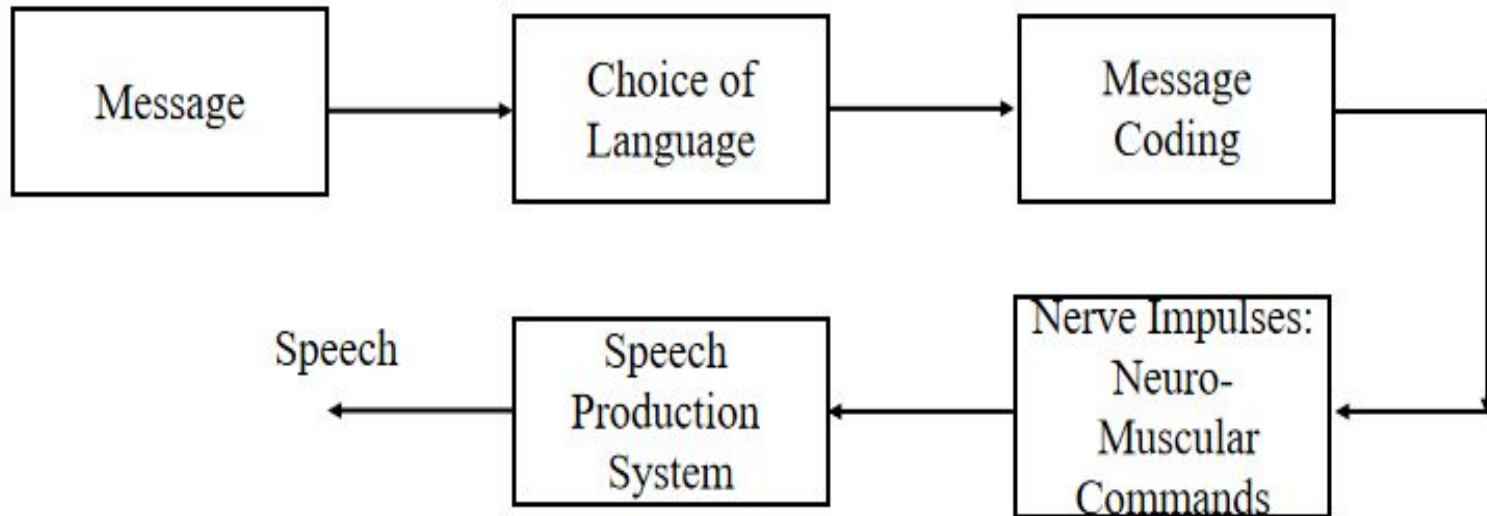
Electrical:



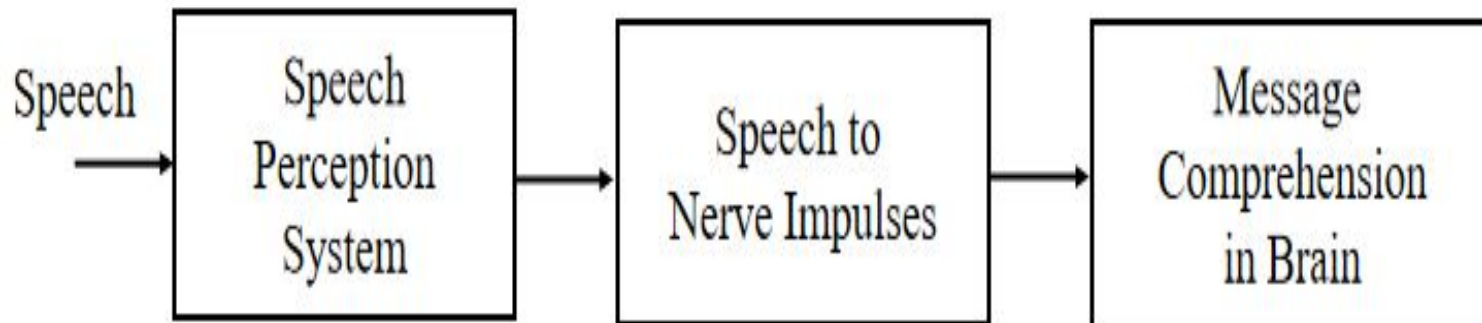
- **Talker:** Message formulation and conveying via speech mode.
- **Listener:** Reception of speech and message comprehension.
- **Medium:** Physical medium which carries speech from talker to listener.

Speech Chain: Steps in Human Speech Communication

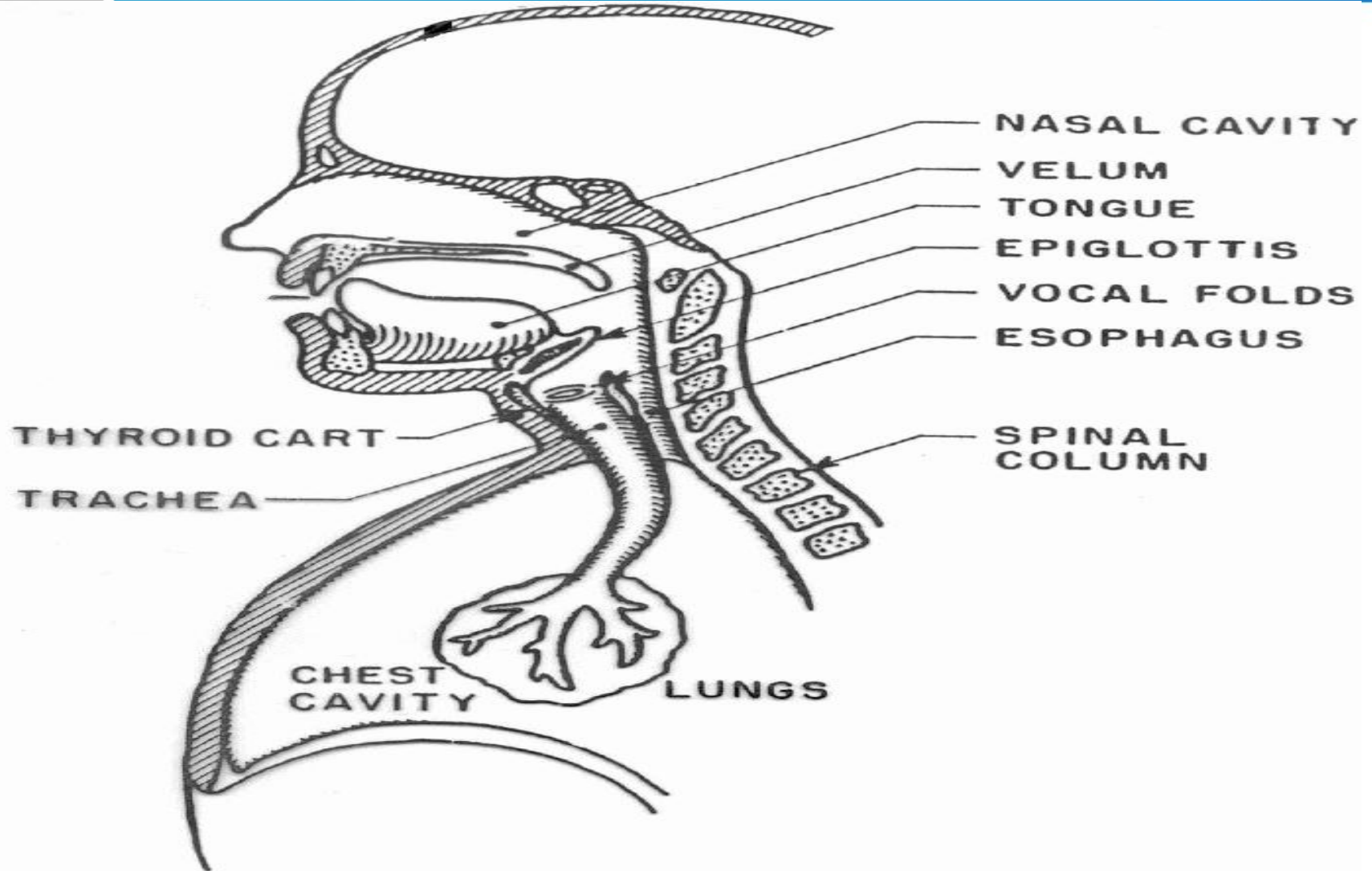
Talker:



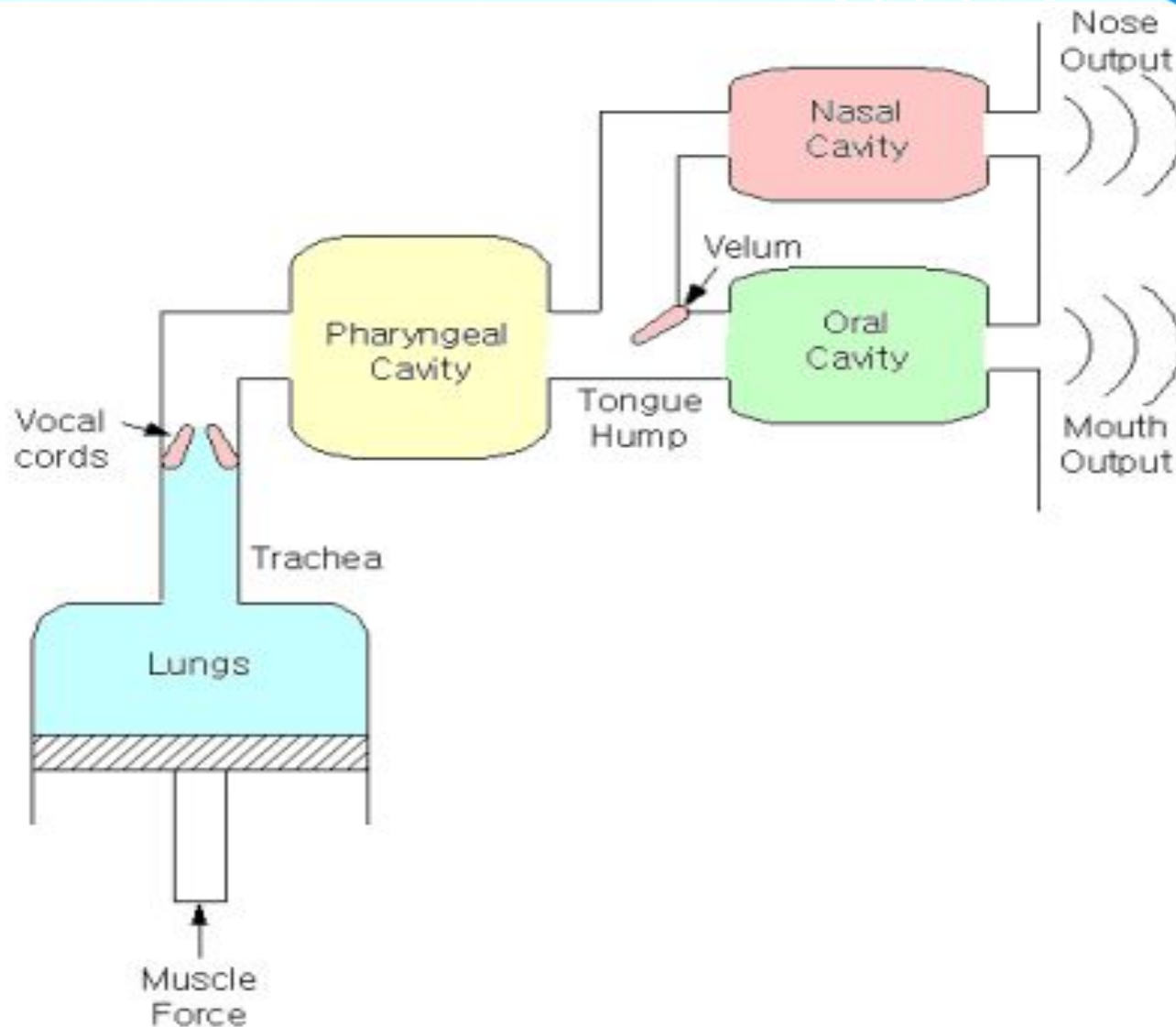
Listener:



Speech Production System



Schematic Representation of Speech Production Process



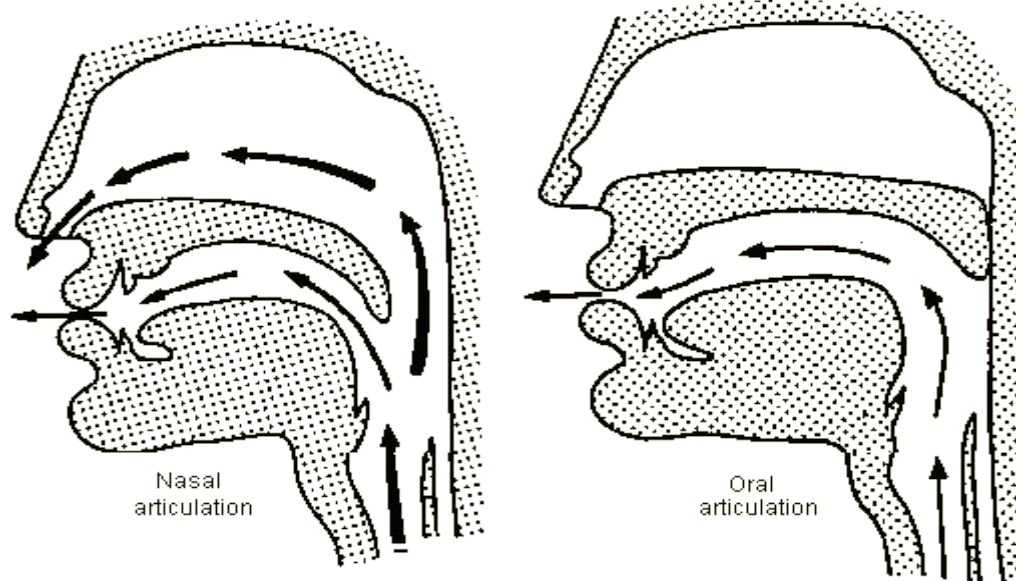
- **Lungs (Air Source):** Lungs provide the airflow (air pressure) required for speech production.
 - **Muscle force** from the diaphragm pushes air upward through the trachea.
- **Vocal Cords (Sound Source):** Air passes through the vocal cords (vocal folds) in the larynx.
 - Vibrating vocal cords produce voiced sounds (e.g., vowels), while open vocal cords produce unvoiced sounds (e.g., /s/, /f/).
- **Pharyngeal Cavity:** The sound enters the pharyngeal cavity, which acts as the first resonating chamber and shapes the sound.
- **Oral Cavity:** The tongue, lips, teeth, and jaw modify the airflow and resonance to produce different speech sounds.
 - Most speech sounds are produced through the oral cavity.

- **Velum (Soft Palate):** The velum controls whether air flows through the mouth or the nose.
 - **Velum raised:** Air exits through the mouth, producing oral sounds.
 - **Velum lowered:** Air passes through the nasal cavity, producing nasal sounds such as /m/, /n/, and /ŋ/.
- **Nasal Cavity:** Acts as an additional resonator for nasal sounds before the air exits through the nose.
- **Speech Output:** The shaped sound finally exits through the mouth (oral output) or nose (nasal output) as audible speech.

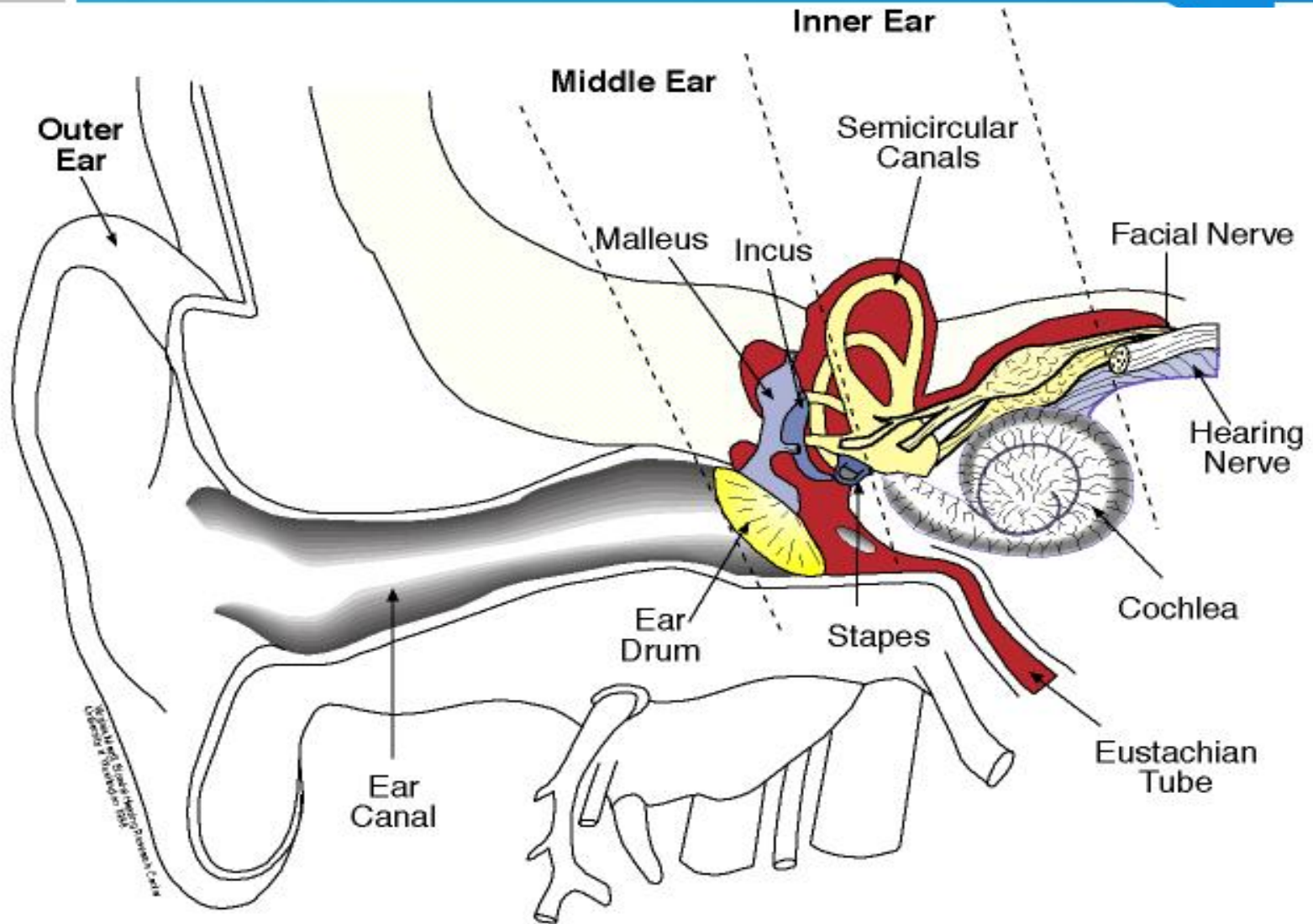
- **Voiced Excitation**
 - Vibration of vocal folds
 - Voiced speech
- **Unvoiced Excitation**
 - Total constriction along the vocal tract
 - Partial constriction along the vocal tract
 - Unvoiced speech
- **Mixed Excitation**
 - Combination of above
 - Mixed speech

Production of Speech Sounds

- **Vowels:** Oral cavity is wide opened, tongue hump, glottal vibration
- **Unvoiced Consonants:** Constriction (Complete closure)
- **Voiced Consonants:** Constriction & glottal vibration
- **Nasal Sounds:** Nasal cavity is coupled
- **Fricatives:** Partial closure, Narrow opening

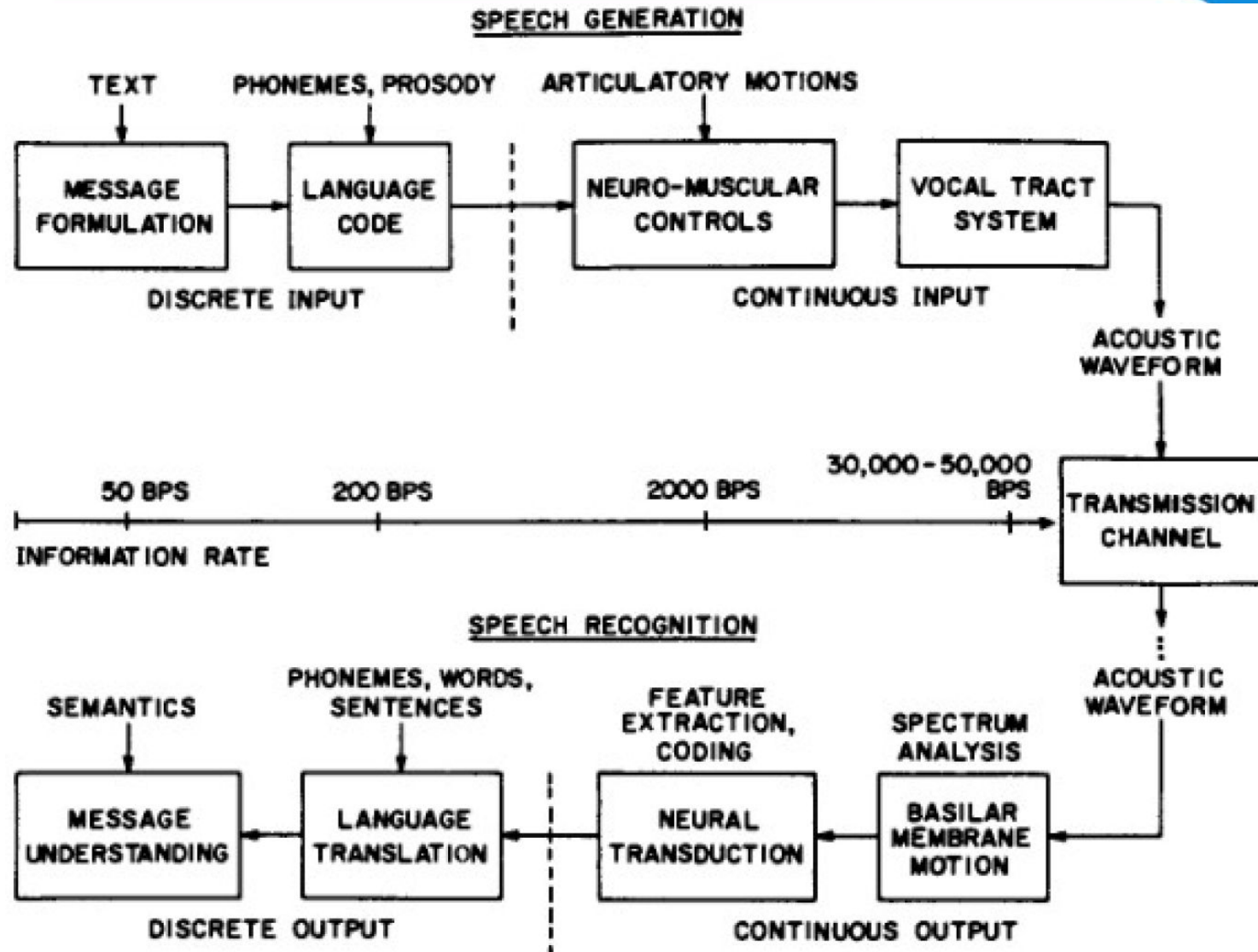


Speech Perception System



- The **speech perception system** consists of three main regions: the outer ear, middle ear, and inner ear.
 - **Outer Ear:** Collects speech sound waves and directs them toward the middle ear.
 - **Middle Ear:** Amplifies and converts sound pressure variations into mechanical vibrations.
 - **Inner Ear:** Converts mechanical vibrations into electrical nerve impulses that are transmitted to the brain via the auditory nerve.
- **Brain:** Higher cognitive centers decode the language and interpret the message to derive meaningful information.
- The audible frequency range for human is appx **20Hz to 20KHz**.

Alternate View of Speech Production and Perception



Acoustic phonetic representation of speech sounds

- **Phonemes:** The number of linguistically distinct speech sounds in a language. Phonemes are often a matter of judgement and is not invariant to different linguists.
- The number of phonemes in a language is not fixed. Different linguists may count them differently based on how they analyze the language.
 - **Example:** In English, some linguists say there are 44 phonemes, while others may count 42, 45, or 46, depending on whether they treat certain sounds as separate phonemes or as variations of the same sound.
- A **phoneme** is the smallest unit of sound that changes the meaning of a word (e.g., /b/ in bat and /p/ in pat). The exact number of phonemes in a language may vary depending on linguistic analysis.

- **IPA (International Phonetic Alphabet)** is a standardized system of symbols used by linguists to represent the sounds of spoken language.
 - It was created by the *International Phonetic Association* to provide a universal way of transcribing speech sounds across all languages.
 - IPA is considered as the global “alphabet of sounds” that makes speech transcription precise and universal.
- **ARPAbet** is a phonetic transcription system developed by ARPA (Advanced Research Projects Agency) for representing speech sounds in computer-readable form. It was widely used in early speech recognition and synthesis systems, especially in the U.S.
 - **It is ASCII-based.** Uses plain text characters instead of IPA symbols, making it easy for computers to process.

Phonetic list for American English

TABLE 2.1. A condensed list of phonetic symbols for American English.

Phoneme	ARPABET	Example	Phoneme	ARPABET	Example
/ɪ/	IY	<u>beat</u>	/ŋ/	NX	<u>sing</u>
/ʊ/	IH	<u>bit</u>	/p/	P	<u>pet</u>
/e/ (eʲ)	EY	<u>bait</u>	/t/	T	<u>ten</u>
/ɛ/	EH	<u>bet</u>	/k/	K	<u>kit</u>
/æ/	AE	<u>bat</u>	/b/	B	<u>bet</u>
/ɑ/	AA	<u>Bob</u>	/d/	D	<u>debt</u>
/ʌ/	AH	<u>but</u>	/g/	H	<u>get</u>
/ɔ/	AO	<u>bought</u>	/h/	HH	<u>hat</u>
/o/ (oʷ)	OW	<u>boat</u>	/f/	F	<u>fat</u>
/U/	UH	<u>book</u>	/θ/	TH	<u>thing</u>
/u/	UW	<u>boot</u>	/s/	S	<u>sat</u>
/ə/	AX	<u>about</u>	/ʃ/ (sh)	SH	<u>shut</u>
/ɪ/	IX	<u>roses</u>	/v/	V	<u>vat</u>
/ɜ/	ER	<u>bird</u>	/ð/	DH	<u>that</u>
/ə/	AXR	<u>butter</u>	/z/	Z	<u>zoo</u>
/aʷ/	AW	<u>down</u>	/ʒ/ (zh)	ZH	<u>azure</u>
/aʲ/	AY	<u>buy</u>	/tʃ/ (tsh)	CH	<u>church</u>
/ɔʲ/	OY	<u>boy</u>	/j/ (dzh, j)	JH	<u>judge</u>
/y/	Y	<u>you</u>	/w/	WH	<u>which</u>
/w/	W	<u>wit</u>	/l/	EL	<u>battle</u>
/r/	R	<u>rent</u>	/m/	EM	<u>bottom</u>
/l/	L	<u>let</u>	/n/	EN	<u>button</u>
/m/	M	<u>met</u>	/ŋ/	DX	<u>batter</u>
/n/	N	<u>net</u>	/ʔ/	Q	(glottal stop)

Classification of Phonemes

Sec. 2.4 Speech Sounds and Features

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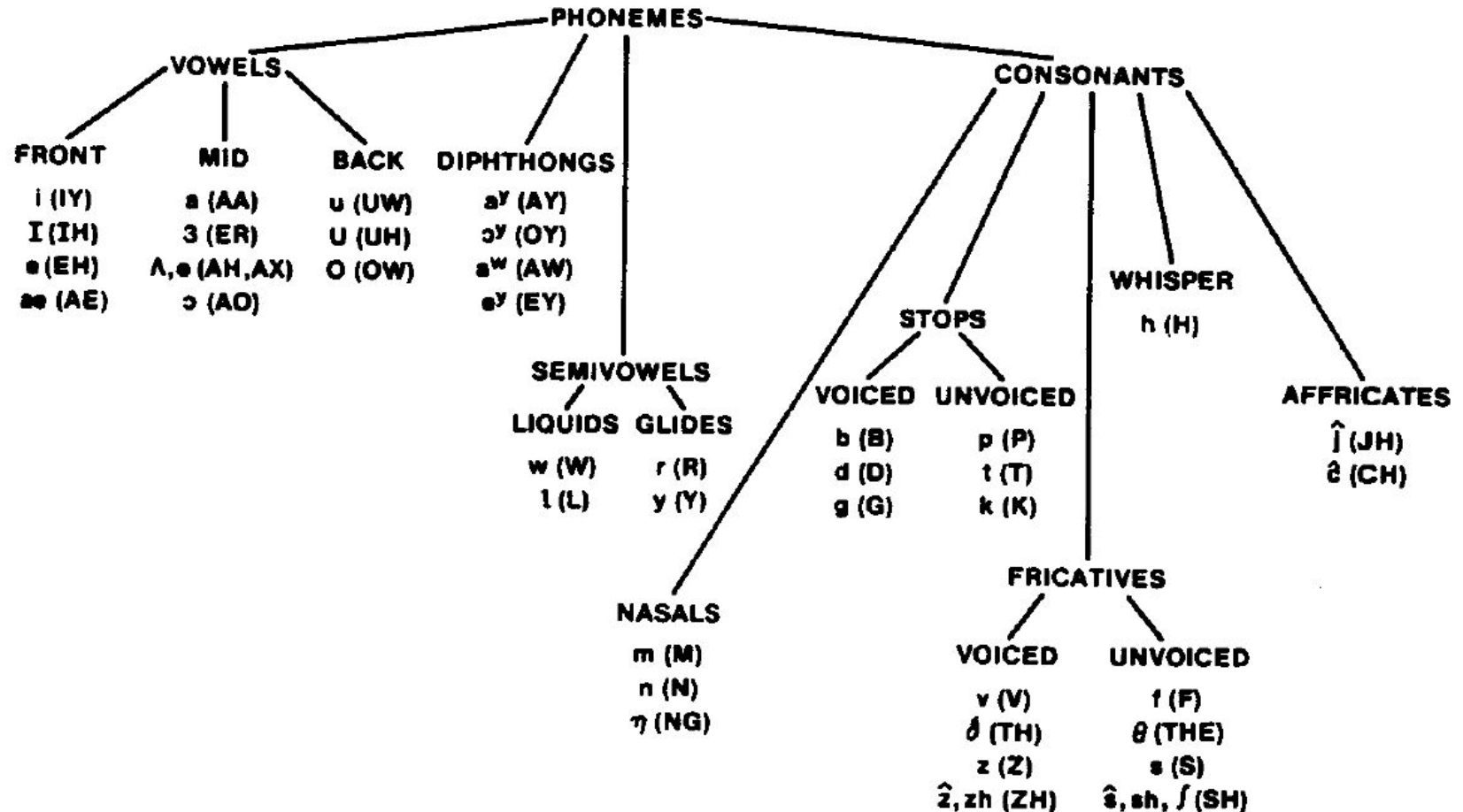


Figure 2.12 Chart of the classification of the standard phonemes of American English into broad sound classes.

		IPA Symbol	ARPAbet (SV)	ARPAbet (UV)	Examples
Vowels	Front	i	i	IY	beat
		I	I	IH	bit
		e	e	EY	bait
		ɛ	E	EH	bet
		æ	@	AE	bat
	Back	ɑ	a	AA	Bob
		ɔ	c	AO	bought
		o	o	OW	boat
		U	U	UH	book
		u	u	UW	boot
	Mid	ɜ	R	ER	bird
		ə	x	AX	ago
		ʌ	A	AH	but
Diphthongs		aI	Y	AY	buy
		aU	W	AW	down
		ɔI	O	OY	boy
		ɪ	X	IX	roses
Semivowels	Liquids	l	l	L	love
		ɫ	L	EL	cattle
		r	r	R	race
	Glides	w	w	W	want
		ʍ	H	WH	when
		j	y	Y	yard

		IPA Symbol	ARPAbet (SV)	ARPAbet (UV)	Examples
Stop Consonants	Voiced	b	b	B	bat
		d	d	D	deep
		g	g	G	go
	Unvoiced	p	p	P	pea
		t	t	T	tea
		k	k	K	kick
Fricatives	Voiced	v	v	V	vice
		ð	D	DH	then
		z	z	Z	zebra
		ʒ	Z	ZH	measure
	Unvoiced	f	f	F	five
		θ	T	TH	thing
Nasal	Non vocalic	s	s	S	so
		ʃ	S	SH	show
	Vocalic	m	m	M	mom
		n	n	N	noon
		ŋ	G	NX	sing
	Vocalic	ɱ	M	EM	some
		ɰ	H	WH	when
		j	y	Y	yard

- Vowels are speech sounds produced with an **open vocal tract**, allowing **air to flow freely**, typically **voiced** and classified by **tongue position** (height and backness), **lip rounding**, and **tenseness**.
- Vowels are generally long in duration compared to consonants.
- They are usually easily and reliably recognized playing significant role for speech recognition, both by human beings and machines.
- **Ex:** “They noted significant improvement in their performance during placement.”
- Th_y n_t_d s_gn_f_c_nt _mpr_v_m_nt _n th__r p_rf_rm_nc_
d_r_ng pl_c_m_nt.
- __e__o__e__i__i__i__a__i__o__e__e__i__ei__e__o__a__e
__u__i__a__e__e__.

Tongue hump and Waveform plots for Vowels

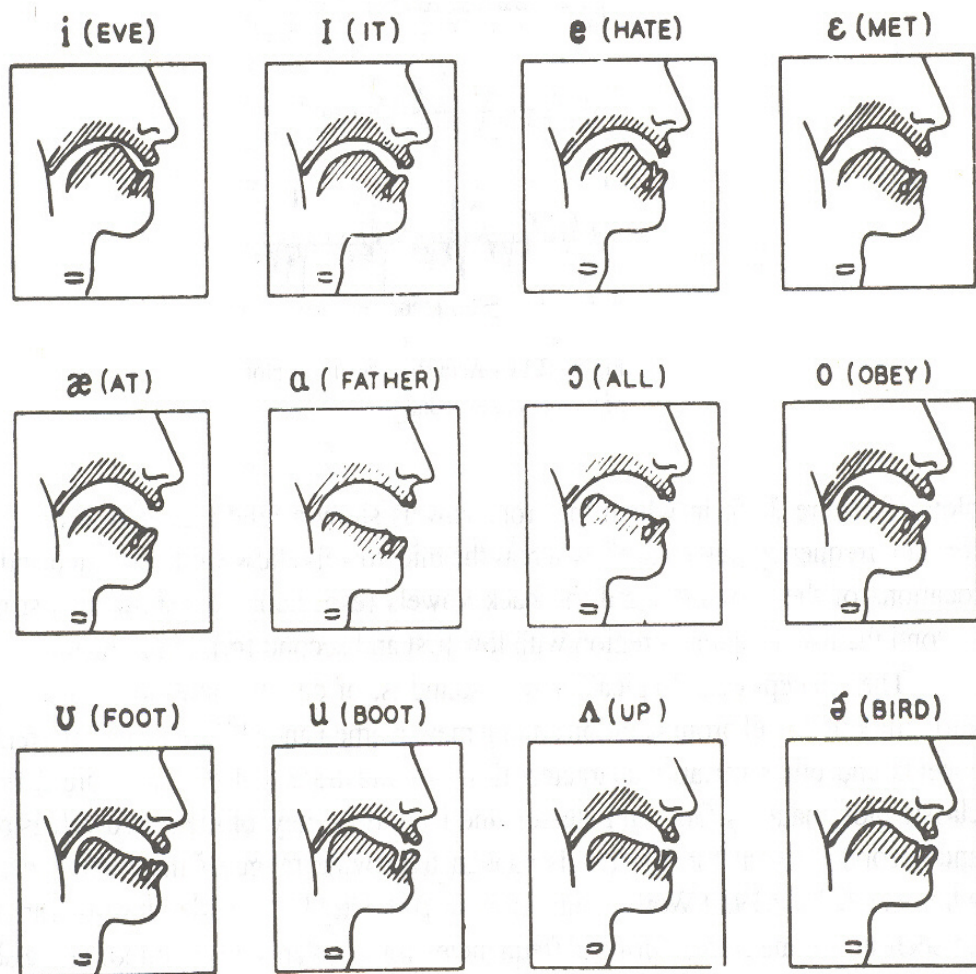


Figure 2.13 Articulatory configurations for typical vowel sounds (after Flanagan [3]).

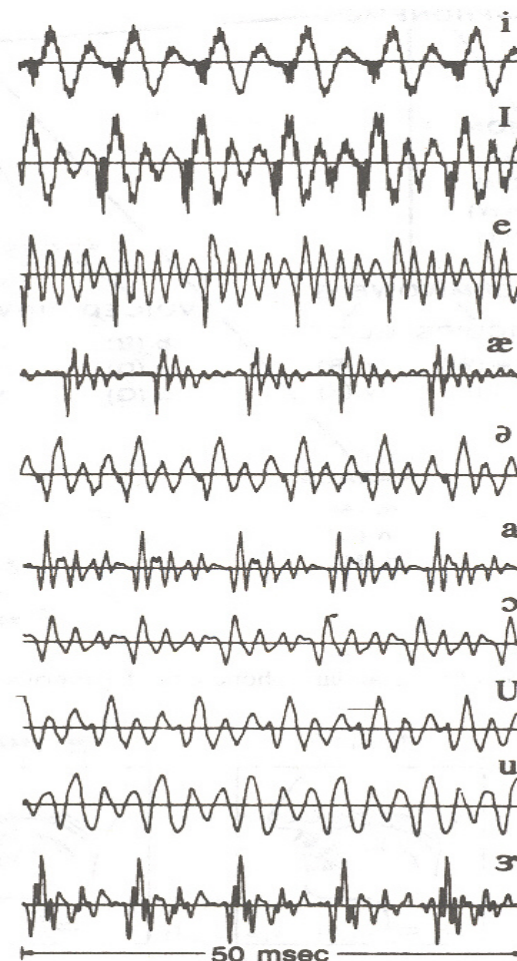


Figure 2.14 Acoustic waveform plots of typical vowel sounds.

Formant Positions for Vowels

- Formant positions are the resonant frequency peaks of the vocal tract that shape vowel quality and distinguish speech sounds.
- The first two formants (**F1 and F2**) are most critical:
 - **F1** corresponds inversely to **tongue height**.
 - **High** tongue position (**close vowels**; /i/, /u/) → **Low F1**
 - **Low** tongue position (**open vowels**; /a/, /æ/) → **High F1**
 - **F2** reflects tongue **frontness/backness**.
 - **Front** vowels (/i/, /e/, /æ/) → **High F2**
 - **Back** vowels (/u/, /o/, /ɔ/) → **Low F2**
- Higher formants (**F3, F4**) add detail to timbre and speaker identity.

F1-F2 Cluster and Centroid for Vowels

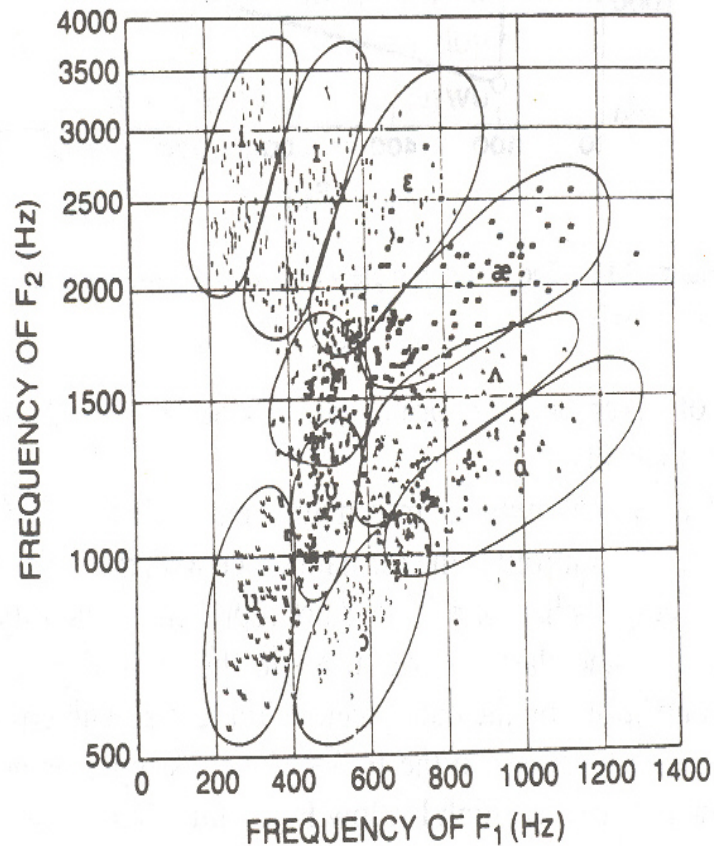


Figure 2.16 Measured frequencies of first and second formants for a wide range of talkers for several vowels (after Peterson & Barney [7]).

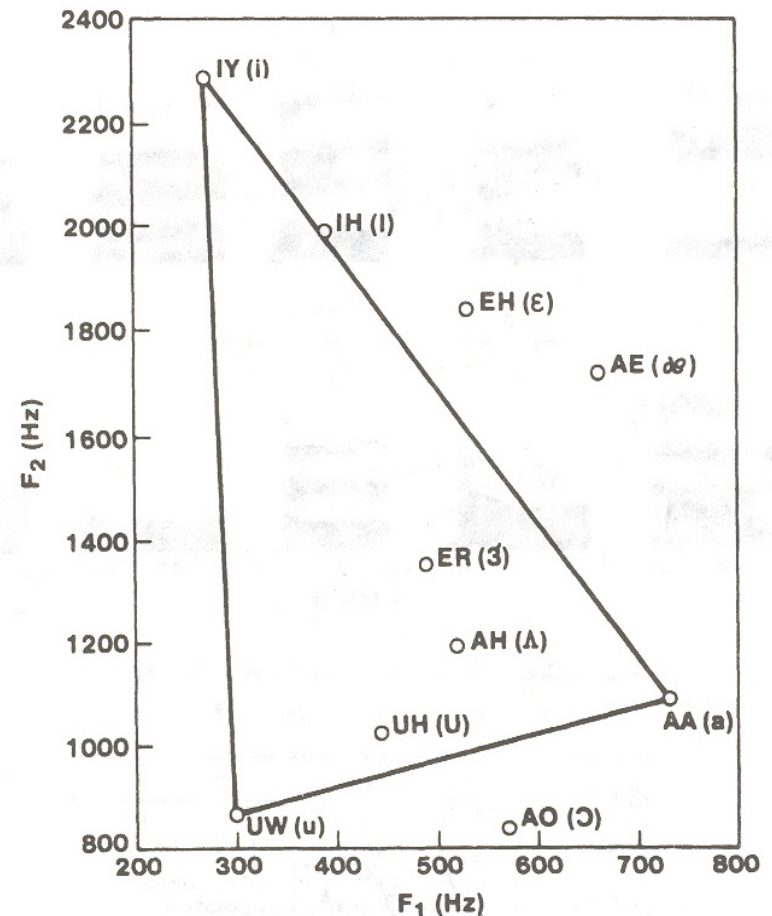


Figure 2.17 The vowel triangle with centroid positions of the common vowels.

- **F1–F2 Cluster:** It is the distribution (grouping) of measured first and second formant frequencies of a vowel from many speakers, showing its natural variation.
 - The F1–F2 cluster shows how the pronunciation of the same vowel varies among different speakers while still being recognized as the same vowel.
- **F1–F2 Centroid:** It is the average (central) F1 and F2 position of a vowel, representing its typical or most representative location in the vowel space.
 - The F1–F2 centroid gives the average position of a vowel, making it useful as a standard reference for vowel identification, comparison, and speech recognition.

Formant frequencies for typical vowels

TABLE 2.2. Formant frequencies for typical vowels.

ARPABET Symbol for Vowel	IPA Symbol	Typical Word	F ₁	F ₂	F ₃
IY	/i/	beet	270	2290	3010
IH	/ɪ/	bit	390	1990	2550
EH	/ɛ/	bet	530	1840	2480
AE	/æ/	bat	660	1720	2410
AH	/ʌ/	but	520	1190	2390
AA	/a/	hot	730	1090	2440
AO	/ɔ/	bought	570	840	2410
UH	/ʊ/	foot	440	1020	2240
UW	/u/	boot	300	870	2240
ER	/ɜ/	bird	490	1350	1690

Diphthongs and Semi-vowels

- A **diphthong** is a *gliding vowel sound* where the tongue moves from one position to another within the same syllable. They start with one vowel and end with another, creating a smooth transition.
- **Examples in English:**
 - /aɪ/ (*bite*) → from /a/ to /ɪ/
 - /eɪ/ (*make*) → from /e/ to /ɪ/
 - /ɔɪ/ (*boy*) → from /ɔ/ to /ɪ/
 - /aʊ/ (*house*) → from /a/ to /ʊ/
 - /oʊ/ (*boat*) → from /o/ to /ʊ/
- **Semi-vowels** (also called **glides**) are sounds that behave like vowels acoustically but function like consonants in syllable structure.
- **Examples in English:**
 - /j/ (*yes*) → similar to vowel /i/
 - /w/ (*we*) → similar to vowel /u/

Consonants, Fricatives, and Stops

- **Nasal Consonants** - Air passes through the nose while the oral cavity is blocked.
- **Ex:** /**m**/ (man) – bilabial nasal; /**n**/ (net) – alveolar nasal
/**ŋ**/ (sing) – velar nasal
- **Fricatives** - Produced by forcing air through a narrow constriction, creating turbulence. Can be voiced or voiceless.
- **Ex: Voiceless:** /**f**/ (fun), /**s**/ (sun), /**ʃ**/ (shoe)
Voiced: /**v**/ (van), /**z**/ (zoo), /**ʒ**/ (measure)
- **Stops (Plosives)** - Complete closure of the vocal tract followed by a sudden release of air. Can be voiced or voiceless.
- **Ex: Voiceless:** /**p**/ (pat), /**t**/ (top), /**k**/ (cat)
Voiced: /**b**/ (bat), /**d**/ (dog), /**g**/ (go)

- **Velar stop consonants:**
 - /ka/, /kha/, /ga/, /gha/
- **Palatal stop consonants:**
 - /ch/, /cha/, /j/, /jha/
- **Alveolar stop consonants:**
 - /T/, /Th/, /D/, /Dh/
- **Dental stop consonants:**
 - /t/, /th/, /d/, /dh/
- **Bilabial stop consonants:**
 - /p/, /ph/, /b/, /bh/